# Joint Estimation of GARCH Option Pricing Model

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**Abstract**

This paper investigates why the Heston-Nandi GARCH model fails to generate realistic implied volatility skew despite successfully fitting return dynamics. Joint estimation using S&P 500 returns and option data from 2007 to 2025 identifies a negative variance risk premium ($\lambda = -0.006$), confirming investors pay for volatility protection. However, the model produces a nearly flat implied volatility smile, with RMSE of 39% compared to market prices. Through counterfactual experiments and analytical derivations, this paper demonstrates that the failure stems from the leverage mechanism itself: the estimated correlation between returns and future variance is only $-0.027$, far below the $-0.3$ to $-0.5$ range required for realistic skew. Even quadrupling the leverage parameter to $\gamma = 20$ yields a correlation of only $-0.09$. The fundamental tension is that GARCH generates skew through gradual leverage effects, while market skew reflects sudden crash risk that leverage cannot capture. This forensic analysis aims at explaining why jump-diffusion extensions are important for option pricing.

# 1 Introduction

GARCH models are the standard tool for modeling time-varying volatility in financial returns (Bollerslev, 1986). When applied to option pricing, these models face a fundamental challenge: parameters estimated from returns alone identify dynamics under the physical measure $\mathbb{P}$, while option prices depend on risk-neutral dynamics under measure $\mathbb{Q}$. The variance risk premium—the compensation investors require for bearing volatility risk—links these measures but cannot be identified from returns alone.

Heston and Nandi (2000) developed a closed-form option pricing framework for GARCH that enables joint estimation using both returns and option prices. This framework has been widely adopted because it allows identification of the variance risk premium while maintaining analytical tractability. However, practitioners and researchers have consistently found that GARCH models struggle to match the steep implied volatility skew observed in equity index options.

This paper takes a different approach than typical model comparisons. Rather than proposing a new model or adding features, the goal is to understand mechanically *why* the Heston-Nandi GARCH model fails for option pricing. Through analytical derivations and counterfactual simulations, the paper provides a forensic analysis of the model's limitations.

The central finding is that GARCH models generate implied volatility skew through the leverage effect—the tendency for volatility to rise after negative returns. However, the leverage effect operates gradually: a negative return today increases variance tomorrow by a small amount. Market option prices, in contrast, reflect fear of sudden crashes. The steep skew in short-maturity option prices appears to reflect crash protection that the gradual leverage mechanism may struggle to replicate fully."

The paper proceeds as follows. Section 2 presents the Heston-Nandi model specification with particular attention to the leverage mechanism. Section 3 describes the data. Section 4 explains the estimation methodology. Section 5 reports parameter estimates and model fit. Section 6 provides the forensic analysis, demonstrating through counterfactual experiments exactly why the model fails. Section 7 concludes.

# 2 Model Specification

## 2.1 The Heston-Nandi GARCH Model

The Heston-Nandi GARCH model (Heston and Nandi, 2000) specifies log-returns under the physical measure $\mathbb{P}$ as:

$$r_t = \log(S_t/S_{t-1}) = \mu + \lambda h_t + \sqrt{h_t}\, z_t \tag{1}$$

where $z_t \sim N(0,1)$ and the conditional variance follows:

$$h_{t+1} = \omega + \beta h_t + \alpha(z_t - \gamma\sqrt{h_t})^2 \tag{2}$$

The parameters have the following interpretation. The parameter $\mu$ represents the daily mean return, while $\lambda$ captures the price of variance risk. The variance intercept $\omega$ determines the long-run variance level, and $\beta$ measures variance persistence. The parameter $\alpha$ quantifies the impact of shocks on variance, and $\gamma$ is the leverage parameter controlling asymmetry.

## 2.2 The Leverage Mechanism

The leverage effect is the key mechanism by which GARCH models generate skew. To understand how it works, expand equation (2):

$$h_{t+1} = \omega + \beta h_t + \alpha z_t^2 - 2\alpha\gamma\sqrt{h_t} z_t + \alpha\gamma^2 h_t \tag{3}$$

The term $-2\alpha\gamma\sqrt{h_t}z_t$ creates asymmetry. When $z_t < 0$ (negative return shock) and $\gamma > 0$, this term is positive, adding to variance. When $z_t > 0$ (positive return shock), this term is negative, reducing variance. This captures the empirical observation that negative returns increase volatility more than positive returns of equal magnitude (Black, 1976).

The covariance between returns and future variance is:

$$\mathrm{Cov}(r_t, h_{t+1}) = \mathrm{Cov}(\sqrt{h_t}z_t, -2\alpha\gamma\sqrt{h_t}z_t) = -2\alpha\gamma\bar{h} \tag{4}$$

where $\bar{h}$ is the unconditional variance. This derivation reveals that the leverage effect strength is proportional to the product $\alpha\gamma$, not to $\gamma$ alone. This observation is central to understanding why the model fails.

Stationarity requires $\beta + \alpha\gamma^2 < 1$, and the unconditional variance is:

$$\bar{h} = \frac{\omega + \alpha}{1 - \beta - \alpha\gamma^2} \tag{5}$$

## 2.3 Risk-Neutral Dynamics

For option pricing, dynamics under the risk-neutral measure $\mathbb{Q}$ are required. The transformation involves shifting the leverage parameter:

$$\gamma^* = \gamma + \lambda + \frac{1}{2} \tag{6}$$

Under $\mathbb{Q}$, the return process becomes:

$$r_t = r_f - \frac{1}{2}h_t + \sqrt{h_t}\, z_t^* \tag{7}$$

with variance dynamics:

$$h_{t+1} = \omega + \beta h_t + \alpha(z_t^* - \gamma^* \sqrt{h_t})^2 \tag{8}$$

The parameter $\lambda$ represents the variance risk premium. When $\lambda < 0$, investors accept lower expected returns for assets that pay off when variance is high, indicating a preference for variance protection. This parameter cannot be identified from returns alone—option prices are required.

### 2.4 Option Pricing

The Heston-Nandi framework permits closed-form option pricing via the characteristic function. European option prices are computed using the Fast Fourier Transform (FFT) method of Carr and Madan (1999). Model-implied volatilities are then obtained by inverting the Black-Scholes formula. Details are provided in Appendix A.

## 3 Data

Daily S&P 500 index prices are obtained from Yahoo Finance for the period January 2007 to May 2025, yielding 4,395 trading days. Log-returns are computed as $r_t = \log(S_t/S_{t-1})$. The sample covers multiple volatility regimes, including the 2008 financial crisis and the 2020 COVID-19 shock. Returns exhibit the standard stylized facts: near-zero mean (0.04%), negative skewness ($-0.41$), and substantial excess kurtosis (12.8). Summary statistics are reported in Table 1 in the Appendix.

Option-implied volatility data for S&P 500 index options span December 2007 to May 2025. Moneyness is defined as $K/S \times 100$, with the dataset including levels from 80% to 120%. Maturities range from one month to twelve months. For estimation, a subset focusing on ATM and OTM calls (100%–120% moneyness), maturities of 1M, 3M, and 6M, and approximately monthly observations is used, yielding 3,312 IV observations.

Returns and IV data are aligned by date, with the risk-free rate set to 2% annually.

## 4 Estimation Methodology

### 4.1 Returns-Only Estimation

Parameters are first estimated using only return data via maximum likelihood. Under the assumption that standardized residuals are i.i.d. standard normal, the log-likelihood is:

$$\mathcal{L}_{returns}(\theta) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\left[\log(h_t) + \frac{(r_t - \mu - \lambda h_t)^2}{h_t}\right] \tag{9}$$

A key limitation is that the variance risk premium $\lambda$ is not separately identified from

$\mu$—only their combination affects the likelihood. Following standard practice, $\lambda$ is set to zero for returns-only estimation.

### 4.2 Joint Estimation

Joint estimation combines return data with implied volatility data. The objective function is:

$$\mathcal{Q}(\theta) = -\mathcal{L}_{returns}(\theta) + w_{IV} \cdot \sum_{i=1}^{N_{IV}} \left(\sigma_i^{market} - \sigma_i^{model}(\theta)\right)^2 \tag{10}$$

The weight $w_{IV} = 1000$ balances the two components. Model-implied volatilities are computed by first obtaining option prices via FFT, then inverting the Black-Scholes formula.

Optimization uses the L-BFGS-B algorithm with box constraints. The bounds on $\gamma$ are set to $[-5, 5]$ based on values reported in the literature (Heston and Nandi, 2000; Christoffersen et al., 2013). Wider bounds lead to extreme estimates that produce numerically unstable option prices.

## 5 Results

### 5.1 Parameter Estimates

Table 2 in the Appendix presents parameter estimates under both estimation approaches. Several features are noteworthy. First, the variance risk premium $\lambda$ is identified as negative ($-0.006$) under joint estimation, consistent with investors paying a premium for variance protection. Second, the leverage parameter $\gamma$ hits its upper bound of 5, indicating the optimizer seeks stronger asymmetry than constraints permit. Third, and most critically, the product $\alpha\gamma = 3.66 \times 10^{-5}$ is extremely small. As shown in equation (4), this product determines leverage strength, not $\gamma$ alone.

### 5.2 Implied Volatility Fit

Table 3 in the Appendix summarizes implied volatility fit. The fit is poor. ATM options have acceptable RMSE (12%), but errors increase dramatically for OTM options, reaching 50%. Short-maturity options show the worst fit (57% for 1M), improving for longer maturities. This pattern—worst fit for OTM short-maturity options—is the signature of missing jump risk.

## 6 Forensic Analysis: Why the Model Fails

The previous section documented that the model fails. This section explains *why* it fails through analytical derivations and counterfactual experiments. The analysis tries to show that it is a failure of the GARCH leverage mechanism, not a consequence of estimation choices.

## 6.1 The Leverage Effect is Too Weak

The leverage effect can be measured by the correlation between returns and future variance:

$$\rho = \mathrm{Corr}(r_t, h_{t+1}) \tag{11}$$

For realistic equity skew, this correlation should be strongly negative, typically in the range $-0.3$ to $-0.5$ (Bakshi et al., 1997; Bates, 2000). Monte Carlo simulation with the estimated parameters yields $\rho = -0.027$—an order of magnitude weaker than required.

Figure 1 shows how this correlation varies with the leverage parameter $\gamma$, holding other parameters fixed at their estimated values. Even at $\gamma = 20$ (four times the constrained estimate), the correlation reaches only $-0.09$—still far below the $-0.3$ threshold needed for realistic skew.

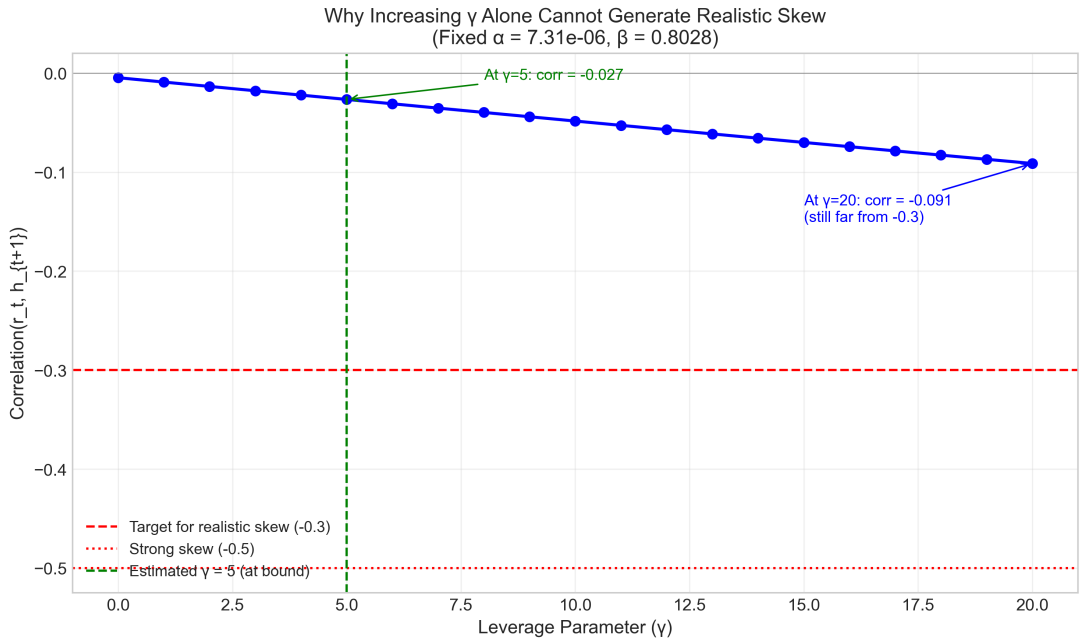


Figure 1: Correlation between returns and future variance as a function of the leverage parameter. Even with $\gamma = 20$, the correlation remains far above the $-0.3$ threshold required for realistic skew.

This experiment reveals that increasing $\gamma$ alone is unlikely to solve the problem. The leverage parameter is already at its bound, and relaxing that constraint would not generate sufficient asymmetry.

## 6.2 The $\alpha$-$\gamma$ Interaction

Equation (4) shows that the covariance between returns and variance is $-2\alpha\gamma\bar{h}$. The leverage strength depends on the *product* $\alpha\gamma$, not on $\gamma$ alone. With $\alpha = 7.31 \times 10^{-6}$ and $\gamma = 5$:

$$\alpha\gamma = 3.66 \times 10^{-5} \tag{12}$$

This product is tiny. To achieve meaningful leverage, both $\alpha$ and $\gamma$ must be large.

Figure 2 shows contours of the return-variance correlation as a function of both parameters. The estimated values (red star) lie in a region where the correlation is only $-0.02$. Achieving the $-0.10$ contour requires either $\gamma > 15$ with current $\alpha$, or $\alpha > 10^{-4}$ (about 15 times larger) with current $\gamma$.

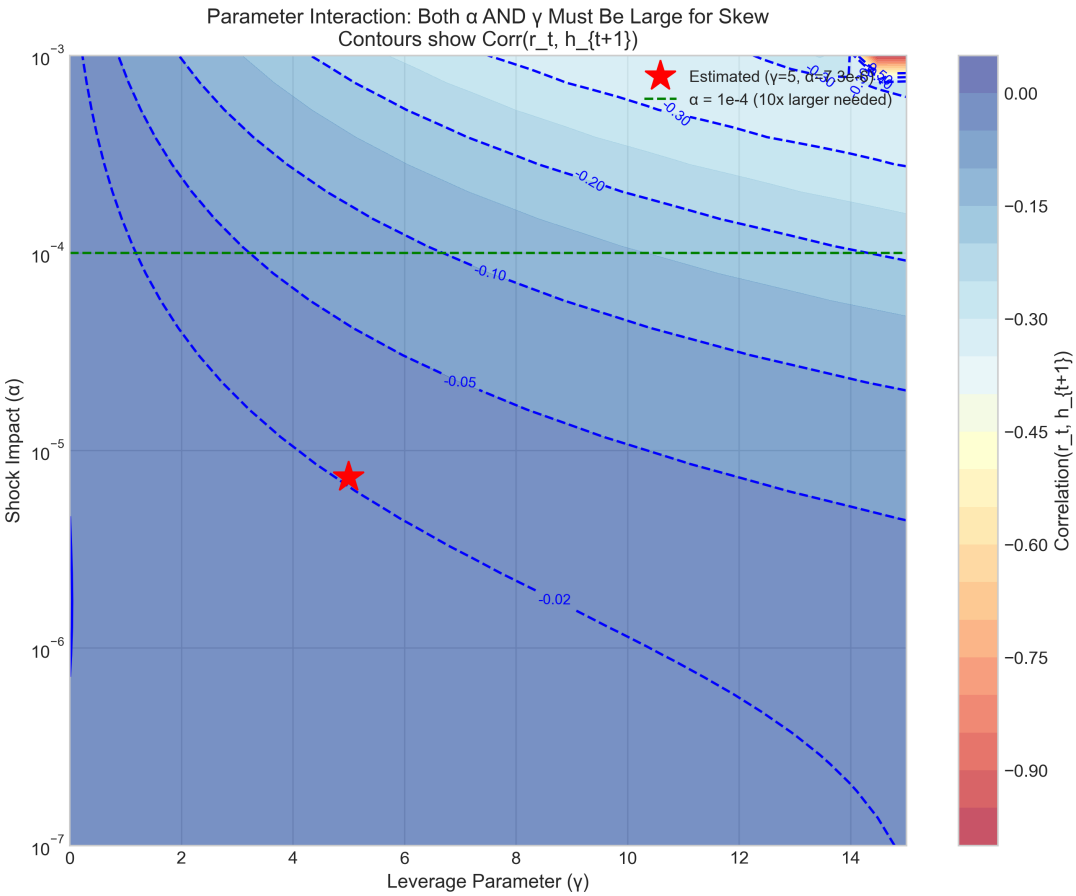


Figure 2: Contours of $\text{Corr}(r_t, h_{t+1})$ as a function of $\alpha$ and $\gamma$. The estimated parameters (red star) lie far from the region that would generate realistic skew. The green dashed line at $\alpha = 10^{-4}$ indicates the approximate threshold needed.

But here is the fundamental problem: increasing $\alpha$ dramatically would worsen the return fit. The parameter $\alpha$ controls how much variance responds to shocks. A value of $\alpha = 10^{-4}$ would make variance 15 times more sensitive to return shocks, creating unrealistic variance dynamics that do not match the data.

This is the core tension: parameters that fit return dynamics (small $\alpha$, moderate $\gamma$) cannot generate sufficient asymmetry for option pricing.

## 6.3 The Variance Ratio Diagnostic

Another way to measure asymmetry is to compare variance following negative versus positive returns. For meaningful skew, variance after negative returns should be 1.5 to 2.0 times higher than after positive returns.

Figure 3 shows this ratio as a function of $\gamma$. At the estimated $\gamma = 5$, the ratio is only 1.02—variance is essentially the same regardless of return sign. Even at $\gamma = 20$, the ratio

reaches only 1.07.

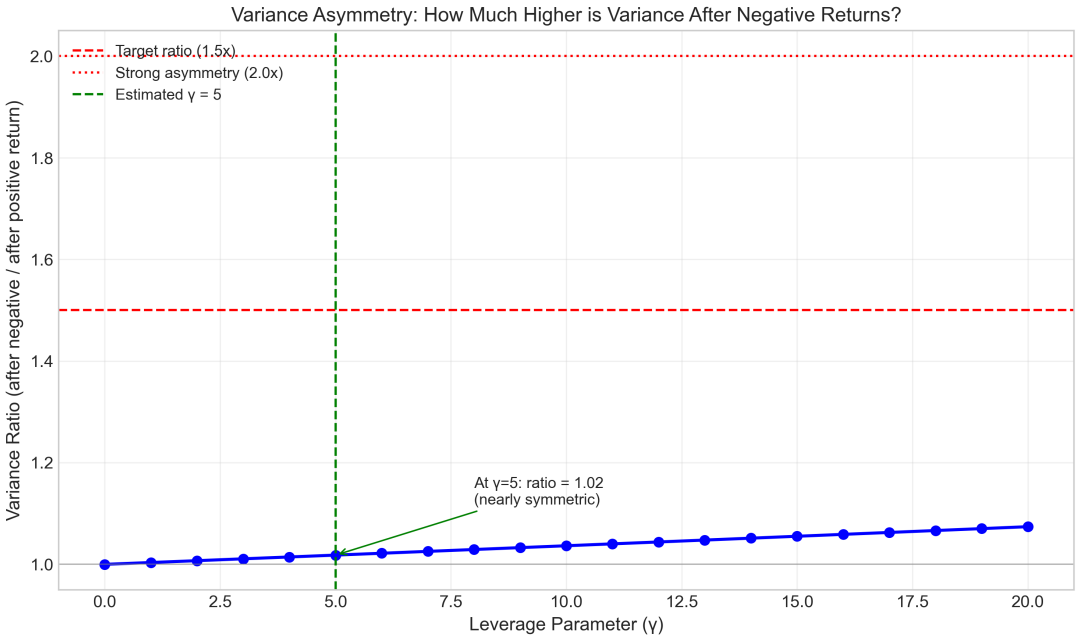


Figure 3: Variance ratio (after negative returns / after positive returns) as a function of $\gamma$. At $\gamma = 5$, the ratio is 1.02—nearly symmetric. A ratio of 1.5–2.0 is needed for realistic skew.

This diagnostic confirms the same conclusion: the model is nearly symmetric. The leverage mechanism is present but far too weak to generate the asymmetry observed in option prices.

## 6.4 The Implied Volatility Smile is Flat

The weak leverage effect translates directly into a flat implied volatility smile. Figure 4 shows model-implied IV smiles for different values of $\gamma$, compared to a typical market smile.

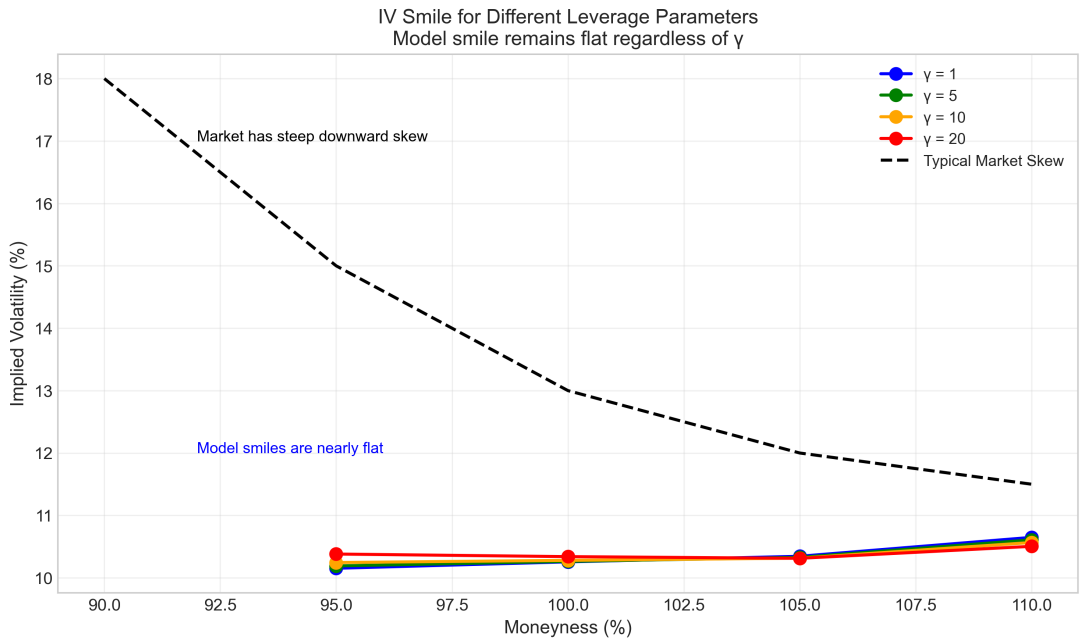


Figure 4: Implied volatility smile for different leverage parameters. All model smiles are nearly flat (10–11%), while the market exhibits steep negative skew (18% for 90% moneyness to 11.5% for 110%).

The market smile shows steep negative skew: IV is about 18% for 90% moneyness (OTM puts) and falls to 11.5% for 110% moneyness (OTM calls). This 6.5 percentage point spread reflects the crash protection premium embedded in put prices.

In contrast, all model smiles are nearly flat, ranging from about 10% to 11% across moneyness levels. Changing $\gamma$ from 1 to 20 barely affects the smile shape. The model cannot generate the realistic asymmetry the market has.

## 6.5 Why Short-Maturity Options Are Difficult to Fit

Figure 5 shows model versus market IV smiles at different maturities. The pattern is revealing: short-maturity options (1M) have the steepest market skew and the largest model error. Longer maturities (6M, 12M) have flatter market skew and smaller errors.

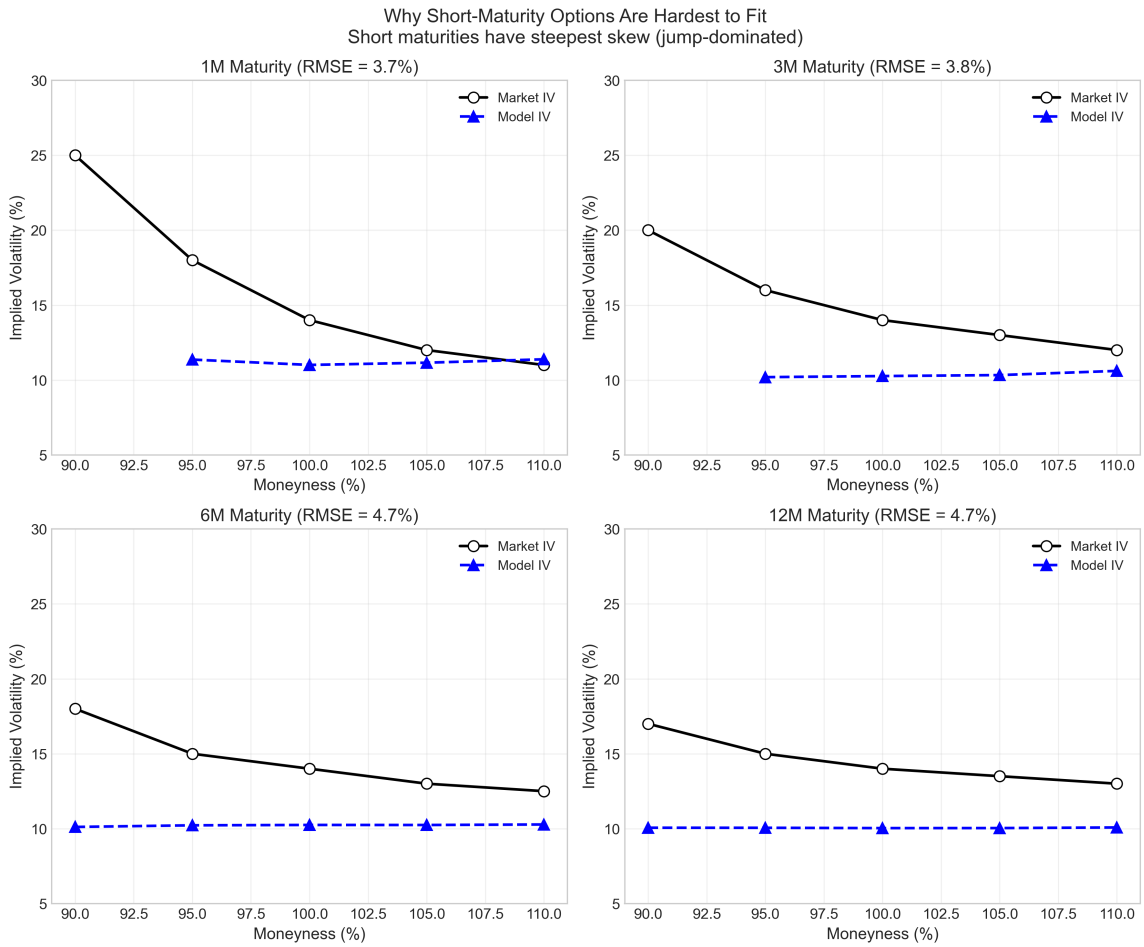


Figure 5: Model versus market IV smile at different maturities. Short maturities have the steepest skew and largest errors. The model produces flat smiles at all maturities.

This pattern has a clear explanation. Short-maturity option prices are dominated by jump risk—the probability of a sudden large move before expiration (Bates, 2000; Pan, 2002). GARCH is a diffusion model: variance evolves gradually, one day at a time. Over a one-month horizon, there is limited time for the leverage effect to accumulate. The steep short-maturity skew prices the risk of a sudden crash, which a diffusion model fundamentally cannot capture.

At longer maturities, the diffusion dynamics have more time to operate, and the market skew flattens (partially). The model fits better, though still poorly, because the mismatch between jump-driven and leverage-driven skew is less severe.

## 6.6 What Parameters Would Generate Realistic Skew?

A natural question is: what parameters *would* generate realistic skew? Simulations indicate that achieving $\mathrm{Corr}(r_t, h_{t+1}) \approx -0.3$ requires approximately:

$$\alpha\gamma \approx 5 \times 10^{-4} \tag{13}$$

With $\gamma = 5$, this implies $\alpha \approx 10^{-4}$—about 15 times larger than the estimated value. Such a parameter would dramatically alter variance dynamics. The Variance would re-

spond about 15 times more strongly to return shocks, leading to a substantial increase in unconditional variance and making variance much more volatile overall.These dynamics would not match observed return behavior. The estimation correctly finds small $\alpha$ because that is what the return data support. The problem is that small $\alpha$ combined with any reasonable $\gamma$ cannot generate skew.

### 6.7 The Fundamental Tension

The forensic analysis reveals a fundamental tension in GARCH option pricing where return dynamics require a small $\alpha$ so that variance responds gradually to shocks, high persistence $\beta$ to capture volatility clustering, and a moderate $\gamma$ to allow for some asymmetry, while option pricing requires a strong leverage effect that needs a large $\alpha\gamma$ product. These requirements are incompatible because parameters that fit returns well with a small $\alpha$ cannot generate the leverage strength needed for option pricing with a large $\alpha\gamma$. This is not a failure of estimation but appears to be a structural limitation of the model.

### 6.8 Why Jumps Would Help

The analysis indicates a solution: jump-diffusion models. Jumps generate skew through a completely different mechanism than leverage. As Bates (2000) and Pan (2002) demonstrate, diffusion-based models cannot match the steep short-maturity skew because leverage effects accumulate only gradually, whereas jump risk is priced immediately, with the leverage mechanism working through negative returns leading to higher variance tomorrow and a slow accumulation of left skewness, while the jump mechanism reflects the possibility of sudden large negative moves that create immediate left skewness in the return distribution. Jump risk is priced independently of the variance risk premium. Even with weak leverage, a negative mean jump size directly creates the left-skewed return distribution that option prices reflect. Importantly, jump effects are strongest at short maturities—exactly where the GARCH model fails most severely. This explains why the literature has moved toward jump-diffusion and stochastic volatility models for option pricing. The forensic analysis in this paper provides the mechanical explanation for why pure GARCH is insufficient.

## 7 Conclusion

This paper provides a forensic analysis of why the Heston-Nandi GARCH model fails to generate realistic implied volatility skew. Joint estimation using S&P 500 data identifies a negative variance risk premium ($\lambda = -0.006$), but the model produces a nearly flat IV smile with 39% RMSE.

The analysis demonstrates that the failure is structural, not a consequence of estimation choices. The leverage mechanism generates skew through the product $\alpha\gamma$, but the estimated $\alpha\gamma = 3.66 \times 10^{-5}$ is too small. The resulting correlation between returns and future variance is only $-0.027$, far below the $-0.3$ to $-0.5$ needed for realistic skew. Counterfactual experiments show that even extreme leverage parameters ($\gamma = 20$) cannot compensate for the small $\alpha$ that return data require.

The fundamental tension is that GARCH generates skew gradually through leverage, while market skew reflects sudden crash risk. Short-maturity options, which have the

steepest skew, are difficult to fit because jump risk dominates at short horizons.

This forensic approach—demonstrating *why* a model fails rather than just documenting that it fails—provides clear motivation for more sophisticated approaches. Jump-diffusion models generate skew through an independent mechanism that does not conflict with return dynamics. The analysis in this paper explains precisely why such extensions are necessary.

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# B Tables

Table 1: Summary Statistics: S&P 500 Daily Log-Returns (2007–2025)

| Statistic | Value |
|---|---|
| Observations | 4,395 |
| Mean (%) | 0.04 |
| Std. Dev. (%) | 1.21 |
| Skewness | −0.41 |
| Excess Kurtosis | 12.8 |
| Minimum (%) | −12.77 |
| Maximum (%) | 8.97 |
| Jarque-Bera | 15,847*** |

*** indicates rejection at 1% level.

Table 2: Heston-Nandi GARCH Parameter Estimates

| Parameter | Returns-Only | Joint | Interpretation |
|---|---|---|---|
| $\mu$ | 0.000455 | 0.000454 | Daily mean return |
| $\omega$ ($\times 10^6$) | 0.721 | 0.721 | Variance intercept |
| $\beta$ | 0.8028 | 0.8045 | Persistence |
| $\alpha$ ($\times 10^6$) | 7.31 | 7.31 | Shock impact |
| $\gamma$ | 5.00 | 5.00 | Leverage (at bound) |
| $\lambda$ | — | −0.006 | Variance risk premium |
| $\alpha\gamma$ ($\times 10^5$) | 3.66 | 3.66 | Leverage strength |
| Persistence | 0.985 | 0.986 | $\beta + \alpha\gamma^2$ |
| Uncond. Vol. (%) | 14.1 | 14.2 | Annualized |

Table 3: Implied Volatility Fit

| Metric | Value |
|---|---|
| Overall RMSE (%) | 39.35 |
| Overall MAE (%) | 31.90 |
| *RMSE by Moneyness:* | |
| 100% (ATM) | 11.73 |
| 105% | 26.72 |
| 110% | 48.75 |
| 115% | 50.38 |
| 120% | 49.40 |
| *RMSE by Maturity:* | |
| 1M | 56.94 |
| 3M | 48.16 |
| 6M | 32.00 |

Table 4: Parameter Bounds

| Parameter | Interpretation | Lower | Upper |
|---|---|---|---|
| $\mu$ | Mean return | $-0.01$ | 0.01 |
| $\omega$ | Variance intercept | $1 \times 10^{-10}$ | $1 \times 10^{-4}$ |
| $\beta$ | Persistence | 0.01 | 0.999 |
| $\alpha$ | Shock impact | $1 \times 10^{-10}$ | $1 \times 10^{-4}$ |
| $\gamma$ | Leverage | $-5$ | 5 |
| $\lambda$ | Variance risk premium | $-0.5$ | 0.5 |

# C Characteristic Function and Option Pricing

The characteristic function for the Heston-Nandi GARCH model under the risk-neutral measure $\mathbb{Q}$ is:

$$\phi(u) = \mathbb{E}^{\mathbb{Q}}\left[e^{iu \cdot x_T} \mid \mathcal{F}_t\right] = \exp\left(A(\tau) + B(\tau)h_{t+1} + iu \cdot x_t\right) \tag{14}$$

where $\tau$ is time to maturity. The coefficients $A(\tau)$ and $B(\tau)$ follow a backward recursion starting from $A(0) = B(0) = 0$:

$$A(t-1) = A(t) + iu \cdot r_f + \omega B(t) - \frac{1}{2}\log(1 - 2\alpha B(t)) \tag{15}$$

$$B(t-1) = iu + \beta B(t) + \frac{(iu + \gamma^* B(t))^2 \alpha}{1 - 2\alpha B(t)} - \frac{1}{2(1 - 2\alpha B(t))} \tag{16}$$

where $\gamma^* = \gamma + \lambda + \frac{1}{2}$.

Option prices are computed via FFT following Carr and Madan (1999):

$$C(K) = \frac{e^{-\alpha_c k}}{\pi} \int_0^\infty e^{-iuk} \psi(u)\, du \tag{17}$$

where $k = \log(K)$, $\alpha_c > 0$ is a dampening parameter, and $\psi(u)$ incorporates the characteristic function.

# D Monte Carlo Validation

Option prices are validated via Monte Carlo simulation under $\mathbb{Q}$. For each path, returns are simulated as $r_t = r_f - \frac{1}{2}h_t + \sqrt{h_t}z_t$ with variance updated as $h_{t+1} = \omega + \beta h_t + \alpha(z_t - \gamma^*\sqrt{h_t})^2$. The option price is the discounted average payoff across 50,000 paths.

# E Analytical Derivation of Leverage Covariance

From the GARCH variance equation:

$$h_{t+1} = \omega + \beta h_t + \alpha(z_t - \gamma\sqrt{h_t})^2 \tag{18}$$

Expanding:

$$h_{t+1} = \omega + \beta h_t + \alpha z_t^2 - 2\alpha\gamma\sqrt{h_t}z_t + \alpha\gamma^2 h_t \tag{19}$$

The return is $r_t = \mu + \lambda h_t + \sqrt{h_t}z_t$. The covariance between $r_t$ and $h_{t+1}$ comes from the cross-term:

$$\begin{aligned}
\mathrm{Cov}(r_t, h_{t+1}) &= \mathrm{Cov}(\sqrt{h_t}z_t, -2\alpha\gamma\sqrt{h_t}z_t) && (20)\\
&= -2\alpha\gamma \cdot E[h_t \cdot z_t^2] && (21)\\
&= -2\alpha\gamma \cdot E[h_t] \cdot E[z_t^2] && (22)\\
&= -2\alpha\gamma\bar{h} && (23)
\end{aligned}$$

This shows that leverage strength is proportional to the product $\alpha\gamma$, explaining why both parameters must be large to generate meaningful asymmetry.